# Deep Learning

CSCI 481

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## Preface

This website collects the lecture notes for CSCI 481[1], "Deep Learning", as taught in Spring Quarter 2027 at Western Washington University[2] by Douglas Ezra Morrison[3].

The course follows the module sequence of Brian Hutchinson[4]'s CSCI 481/581, which he taught from Winter 2021 to Winter 2026.

CSCI 481 builds on DATA 371/471, Machine Learning. Where these notes use an idea that course covers, such as generalization, gradient descent, or logistic regression, they link to its definition in *Machine Learning for Data Science* (lds)[5] rather than defining it again.

The Calendar lists the topics week by week, and the Syllabus gives the course's policies. The main reference text is *Deep Learning* [1], which can be read at no cost on the book's webpage[6].

### References

[1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep learning*. MIT Press, 2016. Available: https://www.deeplearningbook.org/

[1] https://morrison-lab.github.io/swc/course/CSCI-481.html
[2] https://www.wwu.edu/
[3] https://d-morrison.github.io/
[4] https://cs.wwu.edu/hutchib2
[5] https://morrison-lab.github.io/lds/
[6] https://www.deeplearningbook.org/